

Relay event record

Line L0001 CST-001 to CST-030, 138 kV - event of 2026-09-14

Event summary

Protected element	L0001 CST-001 to CST-030 (138 kV, 245 MW rating)
Relay	Line distance relay, terminal CST-001
Trigger	Zone 1 distance element, phase B to ground
Fault inception	2026-09-14 14:02:11.204 CDT
Breaker	CB-112 opened in 3.1 cycles
Reclose	One shot at 30 cycles, unsuccessful; lockout

Sequence of events

- 14:02:11.204 Zone 1 pickup, phase B to ground
- 14:02:11.219 Trip output asserted
- 14:02:11.256 Breaker CB-112 open, current interrupted
- 14:02:11.758 Reclose attempt 1
- 14:02:11.771 Zone 1 pickup on reclose, fault still present
- 14:02:11.802 Breaker CB-112 open, reclosing locked out
- 14:02:12.010 Remote terminal CST-030 open (transfer trip received)

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Fault values

Faulted phase current	6.84 kA
Residual current	6.12 kA
Phase B voltage at fault	0.41 pu
Fault resistance	2.3 ohm (estimated)
Fault location	18.4 km from CST-001 (37 % of line length)

Analysis

The relay operated correctly: a permanent phase B to ground fault inside zone 1.

The unsuccessful reclose confirms the fault was not transient.

Patrol found a damaged insulator string on structure 74, near the computed location.

No misoperation. Recommended: replace the insulator string, then restore the line under switching order SO-2026-0914-01.